

Baris Alparslan

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EDUCATION

Doctor of Philosophy in Statistics (in-progress), *summa cum laude*

August 2023 -

The Ohio State University, Columbus, OH, USA

Relevant Coursework: Statistical Theory, Linear Models, Probability Theory, Measure Theoretic Stochastic Processes, Theory of Causal Inference for Health Science Research, Decision Theory and Bayesian Analysis, Applied Statistical Analysis

Master of Science in Data Science, *summa cum laude*

Jun 2023

Sabanci University, Istanbul, Turkey

Master's Thesis: Monte Carlo Methods for Data Privacy Applications

Advisor: Dr. Sinan Yildirim

Relevant Coursework: Methods of Statistical Inference, Advanced Statistics with R, Machine Learning, Deep Learning, Big Data Processing, Linear Programming

Bachelor of Science in Industrial Engineering, *cum laude*

Jun 2021

Middle East Technical University, Ankara, Turkey

Senior Project: Logistics Network optimization with Capacitated Vehicle Routing

Advisor: Dr. Yasemin Serin

Relevant Coursework: Introduction to Probability, Statistics for Engineers, Introduction to Linear Models, Stochastic Models in Operations Research, Programming with C

PUBLICATIONS

1. **Alparslan, B.**, Yildirim S., Birbil, Ş.İ. Differentially Private Distributed Bayesian Linear Regression with MCMC, **Proceedings of the 40th International Conference on Machine Learning (ICML) 2023**.
(<https://proceedings.mlr.press/v202>)
2. **Alparslan, B.**, Yildirim, S. Statistic Selection and MCMC for Differentially Private Bayesian Estimation, **Statistics and Computing**. 2022, 32, 66 (2022). (<https://doi.org/10.1007/s11222-022-10129-8>)

PROJECTS

Differentially Private Network Analysis

May 2024 – Aug 2025

The Ohio State University, Columbus, OH, USA

- Conducted a literature review on state-of-art approaches on privacy preserving network analysis approaches
- Analyzed convergence properties for Gaussian Mechanism and Randomized Response mechanism for exponential random graph model (ERGM)
- Utilized MATLAB for modeling ERGM networks

Modeling Salmon Spawning Behavior in the Pacific Northwest using MCMC

May 2024

The Ohio State University, Columbus, OH, USA / Bayesian Analysis Course Project

- Developed a Bayesian model to predict salmon spawning behavior using environmental and biological covariates.
- Used RJAGS for Markov Chain Monte Carlo (MCMC) application and orally presented findings with reproducible R scripts and visualizations

TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug 2023 - Present

Department of Statistics, The Ohio State University, Columbus, OH, USA

- Assisted in teaching multiple undergraduate level statistics courses. (Statistics for Business Sciences, Statistical Modelling for Discovery, Introduction to Probability and Statistics for Engineers)
- Conducted weekly recitations and office hours for a class of 70+ students.
- Achieved an average teaching evaluation of 4.56/5. One particular response from a student: "He did a good job explaining all of the lecture materials in recitation in a really detailed way. He always answered any questions anyone had"

Graduate Research and Teaching Assistant

Aug 2021 - Jun 2023

Faculty of Engineering and Natural Sciences, Sabanci University, Istanbul, Turkey

- Contributed to a research project funded by Scientific and Technological Research Council of Turkey (TUBITAK)
- Received a merit-based financial support from TUBITAK in addition to the university stipend
- Provided assistance for various undergraduate level courses (Computational Approaches to Problem Solving, Data Visualization & Analysis, Introduction to Data Science)
- Held weekly recitations and office hours for a class of 30+ students

WORK EXPERIENCE

Statistical Consultant

Aug 2025 - Dec 2025

Statistical Consulting Service, Department of Statistics, The Ohio State University, Columbus, OH, USA

- Provided statistical subject matter expertise to researchers and Principal Investigators on a diverse set of projects across multiple departmental units
- Designed experimental setups and conducted in-depth analysis using R (Quarto), Python and SAS for split-plot designs and general mixed effects models

Data Scientist

Jan 2021 - Jul 2021

KocDigital, Istanbul, Turkey

- Tested and evaluated a predictive maintenance model using Python with Mahalanobis distance model, utilizing Pandas, Numpy and PySpark for statistical anomaly detection and used Hadoop/HDFS for data handling

Data Analyst, Intern

Jun 2020 - Aug 2020

Evreka, Ankara, Turkey

- Developed several performance metrics for cost analysis dashboard in smart waste-management operations
- Represented technical team in meetings with product managers and customers.

COMPUTER SKILLS

R (advanced), Python (advanced), SQL/BigQuery (advanced), MATLAB (intermediate), SAS (intermediate), GAMS & Gurobi (intermediate), Apache Spark (limited experience), C (limited experience), LaTeX, MS Office (Word, PowerPoint, Excel)

AWARDS

University Fellowship

Aug 2023 - Aug 2024

Department of Statistics, The Ohio State University, Columbus, OH, USA

- Received an honorary merit-based fellowship by Department of Statistics recognizing the high potential in research
- Exempted from teaching for a year to focus on coursework and research
- Received a monthly stipend from the department to supporting the research

Full Scholarship

Aug 2021 - Jun 2023

Faculty of Engineering and Natural Sciences, Sabanci University, Istanbul, Turkey

- Full tuition waiver, monthly stipend, and housing awarded by Sabanci University during graduate education.

PROFESSIONAL DEVELOPMENT

International Conference on Machine Learning (ICML)

Aug 2023

Honolulu, HI, USA

- Presented the project "Differentially Private Distributed Bayesian Linear Regression with MCMC"